SESSION 3 OF 8

AI for Requirements & Functional Analysis

DAY 2 — AI Across the SDLC

Duration: 90 Minutes | Hands-on

Anurag Kumar

DAY 2 BEGINS

Day 1 Recap + Where We Are

What we covered on Day 1:

- Session 1: Copilot as a codebase conversationalist
- Session 2: Prompt-driven development + custom instructions
- You can now explain, generate, refactor, and document code

DAY 2

AI Across the SDLC

Today we move beyond code.
Requirements, testing, QA —
the full lifecycle.

Day 1 was about code. Day 2 is about everything around the code.

SESSION GOAL

Translate raw business requirements into structured, detailed specifications

using Copilot

Bridge the gap between business language and engineering artifacts.

WHY THIS SESSION MATTERS

This Session is for Everyone in the Room

DEVELOPERS

You'll translate vague requirements into implementable specs before writing any code. Fewer rewrites, fewer surprises.

FUNCTIONAL ANALYSTS

This is your session. You'll use Copilot to break down requirements, find ambiguities, and create structured artifacts — without writing Java.

QA ENGINEERS

You'll generate test scenarios and edge cases directly from requirements — before any code exists. Shift-left testing, powered by AI.

SESSION AGENDA

90 Minutes | 20 min concept + 60 min hands-on + 10 min discussion

01 **The Requirements Problem**
Why raw requirements fail and how AI helps

02 **User Story Generation**
From one-liner to structured stories with personas

03 **Acceptance Criteria & Edge Cases**
Given-When-Then, failure scenarios, compliance

04 **Domain Modeling with AI**
Sequence flows, entity relationships, API contracts

05 **Cross-Role Collaboration**
Developers, QA, and analysts using the same workspace

THE PROBLEM

Raw Requirements Are Broken

What you receive:

```
"Customer should be able to
transfer funds between
accounts."
```

From: docs/REQUIREMENTS.md

One sentence. No limits, no error handling, no personas, no edge cases.

What's missing:

- What are the transfer limits?
- What if the account is frozen?
- What about insufficient funds?
- Cross-currency transfers allowed?
- Concurrent transfers from same account?
- AML/KYC compliance thresholds?
- Who gets notified? When?

Copilot can help you find what's missing before a single line of code is written.

REQUIREMENTS BREAKDOWN

From One-Liner to Structured Specification

Raw Requirement
Vague, ambiguous, incomplete

→

Copilot Analysis
Identifies gaps, asks the right questions

→

Structured Specification
User stories, criteria, edge cases, flows

What Copilot generates from a single requirement:

15+ user stories	Covering happy path, error scenarios, admin flows
Acceptance criteria	Given-When-Then format for each story
Edge cases	Boundary values, concurrency, compliance scenarios
Sequence flows	End-to-end process from initiation to confirmation

USER STORIES

Generating Stories with Multiple Personas

Standard format: As a [persona], I want [action], so that [benefit]

CUSTOMER	As a customer, I want to transfer funds to another account so that I can pay my bills.
ADMIN	As an admin, I want to view all flagged transfers so that I can review suspicious activity.
SYSTEM	As the system, I want to log every transfer attempt so that we maintain audit compliance.
QA	As a QA engineer, I want to simulate concurrent transfers so that I can verify thread safety.

Copilot generates stories for personas you didn't think of.

ACCEPTANCE CRITERIA

Given-When-Then + Failure Scenarios

Success Scenario

Given Alice has $25,000 in her ACTIVE account

When she transfers $500 to Bob's ACTIVE account

Then Alice's balance is $24,500, Bob's increases by $500, and a transaction record is created

Failure Scenario

Given Alice has $200 in her account

When she tries to transfer $500 to Bob

Then the transfer is rejected with "Insufficient funds", no balances change, and the attempt is audit-logged

Fintech-Specific Edge Cases Copilot Should Surface:

AML threshold ($10K+)	OFAC screening	Daily transfer limits
Cross-currency handling	Concurrent transfers	Account freeze mid-transfer

DOMAIN MODELING

Sequence Flows and API Contracts from Requirements

Ask Copilot to generate the end-to-end flow from a single requirement:

1. User Initiation
2. Authen-tication
3. Account Validation
4. Balance Check
5. Fraud Check
6. Debit & Credit
7. Notifi-cation
8. Audit Log

Copilot can also generate:

- Entity-relationship descriptions from business requirements
- API contract suggestions (endpoints, request/response shapes)
- Domain object identification and their relationships
- State machine definitions (e.g., transaction status transitions)

CROSS-ROLE COLLABORATION

Same Copilot, Different Perspectives

The same requirement generates different artifacts for different roles:

Analyst	**Asks:** Break this requirement into user stories with acceptance criteria	**Gets:** Structured stories, edge cases, ambiguity list
Developer	**Asks:** What technical components are needed to implement this?	**Gets:** Service/controller design, API contracts, DB schema suggestions
QA Engineer	**Asks:** Generate test scenarios for this requirement	**Gets:** Test matrix: functional, security, performance, boundary tests

One requirement, three roles, three sets of artifacts — all from the same Copilot session.

USE CASES

Real-World Scenarios at FIS

ANALYST

One-Liner to User Stories

Take “Customer should be able to transfer funds” and generate 15+ structured user stories covering all personas and scenarios.

QA + ANALYST

Acceptance Criteria

Generate Given-When-Then acceptance criteria for each story, ensuring QA has testable specs from day one.

COMPLIANCE

Regulatory Edge Cases

Ask Copilot to identify AML thresholds, transaction monitoring requirements, and OFAC screening scenarios.

DEVELOPER

Sequence Flow Generation

Generate end-to-end fund transfer flow identifying all services, validations, and external calls involved.

HANDS-ON TIME

60 Minutes of Requirements Analysis with Copilot

Input: docs/REQUIREMENTS.md + Copilot Chat (Ask mode) + Your Domain Knowledge

EXERCISE 1

Generate User Stories from Raw Requirements

What to do:

Open docs/REQUIREMENTS.md. Copy the raw fund transfer requirement. Open Copilot Chat in Ask mode. Ask it to decompose into detailed user stories.

Try this prompt:

```
Break this requirement into detailed user stories
for a fintech system. Include stories for:
- Customer (happy path + errors)
- Admin (monitoring + review)
- System (audit + compliance)
Include failure scenarios.

"Customer should be able to transfer funds
between accounts."
```

Pro Tip: Count the stories. You should get 12–20. If you get fewer than 10, add: “Include edge cases for frozen accounts, daily limits, and concurrent transfers.”

EXERCISE 2

Generate Acceptance Criteria

What to do:

Pick one user story from Exercise 1 (e.g., “As a customer, I want to transfer funds”). Ask Copilot for detailed acceptance criteria.

Try this prompt:

```
Generate acceptance criteria in Given-When-Then
format for this user story:
"As a customer, I want to transfer funds to
another account so that I can pay my bills."

Include at least 5 scenarios:
success, insufficient funds, invalid account,
daily limit exceeded, and system timeout.
```

Pro Tip: For QA: these acceptance criteria become your test cases directly. No translation needed.

EXERCISE 3

Identify Fintech-Specific Edge Cases

What to do:

Ask Copilot to think like a compliance officer. What regulatory and security edge cases exist for the fund transfer feature?

Try this prompt:

```
What are the regulatory and compliance edge
cases for a fund transfer feature in a banking
application?

Consider:
- AML (Anti-Money Laundering) thresholds
- KYC (Know Your Customer) requirements
- OFAC screening
- Transaction monitoring triggers
- PCI-DSS implications
```

Pro Tip: This exercise is gold for functional analysts. Copilot surfaces compliance scenarios that often get discovered late in the project.

EXERCISE 4

Generate a Sequence Flow

What to do:

Ask Copilot to produce an end-to-end sequence flow for the fund transfer process, showing every step from user initiation to confirmation.

Try this prompt:

```
Create a detailed sequence flow for the fund
transfer process showing these steps:
user initiation, authentication, account
validation, balance check, fraud check, debit,
credit, notification, and audit logging.

For each step, note which service handles it
and what happens on failure.
```

Pro Tip: Ask the follow-up: “Convert this to a Mermaid sequence diagram.” Copilot can generate diagram code you can render.

EXERCISE 5

Break Down by Role

What to do:

Take a complex requirement and ask Copilot to assign tasks to different team roles. This shows how AI helps with sprint planning.

Try this prompt:

```
Break down this requirement into tasks for three roles: developer, QA engineer, and
functional analyst:

"Implement real-time fraud detection for transactions above $10,000."

For each role, list:
- Specific deliverables
- Estimated complexity (S/M/L)
- Dependencies on other roles
```

Pro Tip: This is a sprint planning exercise. Show your lead how Copilot can help decompose features across the team.

PROMPT REFERENCE

Session 3 — All Key Prompts

1. Break this requirement into detailed user stories for a fintech system. Include failure scenarios.

2. Generate acceptance criteria in Given-When-Then format. Include 5+ scenarios: success, failures, edge cases.

3. What are the regulatory and compliance edge cases? Consider AML, KYC, OFAC, transaction monitoring.

4. Create a sequence flow for the fund transfer process. Note which service handles each step.

5. Break down this requirement into tasks for developer, QA, and analyst. List deliverables and complexity.

CORE CONCEPT + WRAP-UP

AI-Assisted Product Thinking

AI doesn’t replace business analysis — it accelerates it.
Copilot helps you think of scenarios you might have missed.

Reflection

- How many edge cases did Copilot surface that you hadn’t considered?
- Could your team use this to speed up sprint planning?
- How would this change the handoff between analysts and developers?

UP NEXT

Session 4

AI-Powered Testing
& QA Automation